

INTERNAL ANALYSIS · APRIL 2026

PV + BESS Scenario Analysis

Softades & Αρκατζιά

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

PV: €0.72/Wp

BESS: €127k/MWh

Tariff: €0.16/kWh

LTV: 65% @ 5.5% / 12yr

Albedo: 0.70 white ground

PANEL SPEC

AE Solar Meteor

Flttype TOPCon, Eff_{cell} 24.5%, Eff_{mod} 23.5%

Eff_{cell} 24.5%, Eff_{mod} 23.5%

30yr product warranty

EPC & EQUIPMENT

PV EPC

€720/kWp

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

BESS 4h

€127k/MWh

LFP installed

PERMITTING

€35,000

Permitting

SOUTH YIELD

2,150 kWh/kWp

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

EW YIELD

1,450 kWh/kWp

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

PV O&M

€15/kWp

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

BESS O&M

€2,500/MWh

Permitting costs, LFP installed, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

FIXED OVERHEAD

€2,000–2,500

AE Solar Meteor bifacial, White albedo, 4h BESS, Bank financing, DSM — Lighthief confirmed costs

LTV

65%

Senior debt

RATE

5.5% p.a.

Fixed / green loan

TERM

12 years

Annuity factor 0.116

1

Παρασόλια, Σοφτάδες — Λάρνακα

Plot 0/2150 · Παύλος Παύλου · Agricultural · 1,363 m²

* South-Facing · 25°

95 kWp · 44 panels

CAPITAL EXPENDITURE

• PV EPC (95 kWp × €720)

€68,400

• BESS 4h — 380 kWh

€48,260

0.38 MWh × €127,000/MWh

• Permitting & Docs

€35,000

TOTAL CAPEX

€151,660

ANNUAL PERFORMANCE

PRODUCTION

204 MWh

GROSS REVENUE

€32,640

O&M COST/YR

-€4,375

NET EBITDA

€28,265

BANK FINANCING

Loan (65%) @ 5.5% / 12yr

€98,579

Equity required (35%)

€53,081

Annual debt service

-€11,435

NET CASH / YR

€16,830

DSCR 2.86x

Unlevered payback

5.4 yr

Equity payback (levered)

3.2 yr

↔ East-West · 12°

150 kWp · 69 panels

CAPITAL EXPENDITURE

• PV EPC (150 kWp × €720)

€108,000

• BESS 4h — 600 kWh

€76,200

0.60 MWh × €127,000/MWh

• Permitting & Docs

€35,000

TOTAL CAPEX

€219,200

ANNUAL PERFORMANCE

PRODUCTION

218 MWh

GROSS REVENUE

€34,800

O&M COST/YR

-€5,750

NET EBITDA

€29,050

BANK FINANCING

Loan (65%) @ 5.5% / 12yr

€142,480

Equity required (35%)

€76,720

Annual debt service

-€16,528

NET CASH / YR

€6,522

DSCR 2.11x

Unlevered payback

7.5 yr

Equity payback (levered)

11.8 yr

2

Αρκατζιά, Δήμος Κώτηρας — Αμμόχωστος

Plot 0/2257 · Παύλου Κούλλα · Χωράφι · Provisional title

* South-Facing · 25°

270 kWp · 125 panels

CAPITAL EXPENDITURE

• PV EPC (270 kWp × €720)

€194,400

• BESS 4h — 1,080 kWh

€137,160

1.08 MWh × €127,000/MWh

• Permitting & Docs

€35,000

TOTAL CAPEX

€366,560

ANNUAL PERFORMANCE

PRODUCTION

581 MWh

GROSS REVENUE

€92,880

O&M COST/YR

-€9,250

NET EBITDA

€83,630

↔ East-West · 12°

420 kWp · 194 panels

CAPITAL EXPENDITURE

• PV EPC (420 kWp × €720)

€302,400

• BESS 4h — 1,680 kWh

€213,360

1.68 MWh × €127,000/MWh

• Permitting & Docs

€35,000

TOTAL CAPEX

€550,760

ANNUAL PERFORMANCE

PRODUCTION

609 MWh

GROSS REVENUE

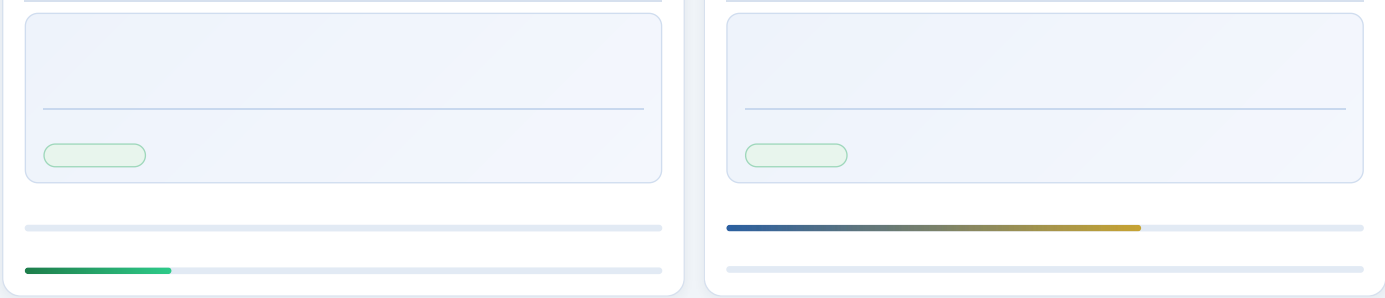
€97,440

O&M COST/YR

-€13,000

NET EBITDA

€84,440



O&M Breakdown — All Scenarios

PLOT 1 · SOUTH · 95 KWP		PLOT 1 · EW · 150 KWP		PLOT 2 · SOUTH · 270 KWP		PLOT 2 · EW · 420 KWP	
PV O&M (95 × €15)	€1,425	PV O&M (150 × €15)	€2,250	PV O&M (270 × €15)	€4,050	PV O&M (420 × €15)	€6,300
BESS O&M (0.38 MWh × €2,500)	€950	BESS O&M (0.60 MWh × €2,500)	€1,500	BESS O&M (1.08 MWh × €2,500)	€2,700	BESS O&M (1.68 MWh × €2,500)	€4,200
Monitoring + insurance	€2,000	Monitoring + insurance	€2,000	Monitoring + insurance	€2,500	Monitoring + insurance	€2,500
Total O&M / yr	€4,375	Total O&M / yr	€5,750	Total O&M / yr	€9,250	Total O&M / yr	€13,000

Full Comparison — All Scenarios

SCENARIO	KWP	TOTAL CAPEX	MWH/YR	REVENUE/YR	O&M/YR	EBITDA/YR	EQUITY IN	BEST SVC/YR	NET CASH/YR	PSCR	E
PLOT 1 · SOFATADES · 1,363 M² * South	95	€151,660	204	€32,640	€4,375	€28,265	€53,081	€11,435	€16,830	2.86×	
PLOT 1 · SOFATADES · 1,363 M² ↔ East–West	150	€219,200	218	€34,800	€5,750	€29,050	€76,720	€16,528	€6,522	2.11×	
PLOT 2 · APKATZIA · 3,704 M² * South ★	270	€366,560	581	€92,880	€9,250	€83,630	€128,296	€27,639	€55,991	3.36×	
PLOT 2 · APKATZIA · 3,704 M² ↔ East–West	420	€550,760	609	€97,440	€13,000	€84,440	€192,766	€41,527	€42,913	2.35×	

KEY ASSUMPTIONS & FLAGS

⚠ AE Solar Meteor yield: 2,150 kWh/kWp south (monofacial 1,950 + 10.3% bifacial gain at 80% bifaciality / albedo 0.70); 1,450 kWh/kWp EW (~5% bifacial gain at low tilt). Based on Agios Theodoros RTB confirmed yield methodology.

⚠ BESS sizing: 4h × peak PV kW. All 4 systems (0.38–1.68 MWh) are sub-Linyang minimum (5.015 MWh). Practical BESS would use sub-5MWh commercial units (Huawei/Sungrow). €127k/MWh rate applied proportionally from Agios RTB confirmed stack.

⚠ Bank financing: 65% LTV, 5.5% fixed, 12yr annuity (factor 0.116). Cyprus green/SME loan typical range. Assumes RES license and net metering contract as collateral support.

⚠ O&M: PV €15/kWp/yr + BESS €2,500/MWh/yr + monitoring/insurance €2,000–2,500/yr flat. Based on Agios Theodoros opex stack (pvOm €15.15/kWp, bessOm €2,462/MWh).

⚠ Plot 2 title is Provisional (Προσωρινός τίτλος, Art.10 Law 44/84) — confirm upgrade before permitting. Both plots are agricultural — Town Planning Permission required.